Shreyash Wetal

AI / Machine Learning Engineer • LLMs, RAG, Computer Vision, MLOps
Sydney, NSW | +61 451 507 744 | shreyash.wetal03@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

AI engineer and final-year Master of Computer Science student (AI and Data Science) at the University of Sydney, with industry experience at NVIDIA and The Modern Group. Builds and deploys production AI systems end to end, from conversational LLM and RAG pipelines to real-time computer vision, covering data pipelines, model training, evaluation, cloud deployment and monitoring. Published researcher (AJCAI 2025, Springer) focused on measurable results and clear communication with technical and non-technical teams.

TECHNICAL SKILLS

- **Languages and Engineering:** Python, SQL, JavaScript, Git, REST APIs, Flask, CI/CD, automated testing, code review, technical documentation
- **Generative AI and LLMs:** OpenAI API, LangChain, RAG, embeddings, vector databases, hybrid search (BM25 + dense), prompt engineering, LLM evaluation, MCP
- **ML, Deep Learning and Vision:** TensorFlow, CNNs, transformers, U-Net, DeepLabV3+, YOLOv8, OpenCV, ONNX Runtime, semantic segmentation, NLP, time series
- **Data, Cloud and MLOps:** Pandas, ETL pipelines, data modelling, GeoPandas, Rasterio, AWS, GCP, Firebase, DigitalOcean, Twilio, real-time inference, monitoring
- **Evaluation:** A/B testing, cross-validation, IoU, Dice, F1, ROC-AUC, reproducible benchmarking, responsible AI and privacy

PROFESSIONAL EXPERIENCE

AI Engineer, The Modern Group, Sydney **Jun 2025 – Dec 2025**

- Architected RAG-based conversational AI systems using embeddings and vector retrieval, improving retrieval accuracy by 30% on internal benchmarks.
- Built real-time inference APIs on GCP, Firebase and AWS with CI/CD pipelines, cutting deployment cycle time by 40% and reducing inference latency and cloud costs by 20%.
- Created reusable prompt templates and automation workflows adopted by non-technical teams, translating business requirements into working AI features.

Research Student, Artificial Intelligence, University of Sydney **Mar 2025 – Jul 2026**

- Framed a clinical imaging problem as a segmentation task and trained deep CNNs that improved Dice score by 12% through preprocessing, augmentation and hyperparameter tuning.
- Benchmarked U-Net vs DeepLabV3+ on accuracy, robustness and inference speed with reproducible evaluation (cross-validation, IoU, F1, ROC), documenting trade-offs for clinicians.

Associate Prompt Engineer, NVIDIA, Santa Clara, USA **Oct 2024 – Dec 2024**

- Designed enterprise-grade prompt frameworks that lifted LLM response precision by 22%, adopted by the wider team as its default refinement process.
- Built automated evaluation pipelines scoring LLM outputs on accuracy, clarity and usefulness, and ran A/B tests across transformer models to standardise NLP benchmarking.

PROJECTS

MCP Matchmaking and Discovery Platform | *Python, embeddings, BM25, entity resolution*

- Unified MCP server listings from multiple registries via entity resolution and built hybrid semantic search (dense vectors + BM25 with Reciprocal Rank Fusion) for natural-language tool discovery.
- Added an automated security scanner (permissions, dependencies, secrets, injection risks), shipped as a zero-dependency Python library with 65 automated tests.

AUS Backyard Intelligence | *YOLOv8, GeoPandas, Rasterio, MapLibre GL JS*

- Fine-tuned a YOLO ensemble on aerial imagery to detect pools and solar panels at ∼90% recall with a 5–7% false-positive rate, fused with parcel and terrain data to score properties.
- Built a geospatial backend with compliance logic for all eight Australian jurisdictions and an interactive map front end as a free, open-data alternative to commercial platforms.

Published Research Projects

- **EyeDentify:** deep learning model for non-invasive biometric identification from eye-blink patterns.
- **Smart Drone Surveillance:** integrated UAVs, AI vision and IoT into an automated surveillance system.

EDUCATION

Master of Computer Science (AI and Data Science), University of Sydney **2025 – Dec 2026**
WAM: 76.5 (Distinction)

Bachelor of Technology, Information Technology, Pimpri Chinchwad College of Engineering, India **2020 – 2024**
CGPA: 8.57/10 | FIAT India Academic Excellence Scholarship (2020–2024)

PUBLICATIONS, CERTIFICATIONS AND AWARDS

- **Publications:** AJCAI 2025 (Springer); ICSC 2024; "EyeDentify"; "Smart Drone Surveillance System"; "Intelligent Auto Traffic Signal Controller for Emergency Vehicles" (IJSAR)
- **Certifications:** DeepLearning.AI TensorFlow Developer; Deep Learning Specialisation; NLP in TensorFlow; Time Series; LangChain for LLM Apps
- **Awards:** Mentor, 2025 Humanitarian Innovation Hackathon | Top 20 Finalist, RoboKidz International Robotics Competition